Shivam Kamboj

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Portfolio | GitHub | LinkedIn

Education

University of California, Merced Expected Fall 2026
Ph.D. in Physics; advisor: Prof. Lin Tian
Research: light–matter interactions, quantum Rabi systems, and superconducting-circuit modeling.

San Jose State University 2022
M.S. in Physics

Research Experience

Tian Group, University of California, Merced Merced, CA
Graduate Student Researcher *May 2023 – Present*

- Develop theoretical and computational models of ultrastrong light–matter coupling in the quantum Rabi dimer, connecting Hamiltonian structure to spectra, collective modes, and quantum-state properties.
- Combine normal-mode analysis and sparse exact diagonalization to resolve low-energy states by parity and site-exchange symmetry, distinguish level crossings, and characterize a candidate cat-state doublet.
- Simulate state evolution, entanglement, and dissipation using Python and QuTiP; evaluate convergence with Hilbert-space truncation and run checkpointed parameter studies on Linux clusters with SLURM.
- Build COMSOL electromagnetic models of a proposed superconducting implementation; extract capacitances, mode frequencies, impedances, and inductive participation to assess feasible interactions and device constraints.

Lawrence Livermore National Laboratory Livermore, CA
Physics Graduate Intern, Quantum Coherent Device Physics Group *May – Aug 2024*

- Developed Python software for pulse-level control of superconducting qudits, evaluated gate sequences, and characterized their performance using quantum process tomography.
- Translated experimental measurements into process maps through a generalized linear-inversion tomography pipeline using pseudo-inverse reconstruction.
- Automated Ramsey and T_1/T_2 fitting to characterize device behavior and provide quantitative feedback for control experiments.

Sharping Group, University of California, Merced Merced, CA
Research Collaborator, Experimental Quantum Optics *2024 – Present*

- Developed Hamiltonian models of coupled YIG–cavity systems to study magnon–photon spectra and entanglement, including inter-cavity coupling and parametric driving.

Hurst Group, San Jose State University and UC Merced San Jose / Merced, CA
Graduate Researcher *Jun 2021 – Jan 2024*

- Derived analytical dispersion relations and developed numerical solvers for two-dimensional spin-torque oscillator arrays; studied oscillatory edge modes and robustness to disorder.
- Connected mathematical modeling of driven magnetic systems with numerical studies of edge dynamics, leading to a first-author publication in *Physical Review B*.

Publications and Manuscripts

1. **S. Kamboj**, R. A. Duine, B. Flebus, and H. M. Hurst, "Oscillatory edge modes in two-dimensional spin-torque oscillator arrays," *Physical Review B* **109**, 094436 (2024). doi:10.1103/PhysRevB.109.094436.
2. **S. Kamboj** and L. Tian, "Deterministic Cat States and Parity-Protected Bosonic Qubits in the Quantum Rabi Dimer." Manuscript in preparation.
3. **S. Kamboj**, J. Beurkens, J. Sharping, and L. Tian, "Superconducting realization of the Quantum Rabi Dimer." Manuscript in preparation.

Selected Presentations

- **S. Kamboj** and L. Tian, "The Instability Window: Macroscopic Entanglement and Delayed Symmetry Breaking in the Quantum Rabi Dimer." Quantum Industry Showcase, UC Santa Barbara, 2026.
- **S. Kamboj** and L. Tian, "Adiabatic Evolution and Fock-state Mapping Across the Phase Boundary in a Quantum Rabi Dimer." Poster, APS Global Physics Summit, 2026.
- **S. Kamboj**, Y. Cho, and Y. J. Rosen, "Improving Fidelity of Quantum Gates Using Optimal Control." Poster, PLS Division Student Poster Symposium, Lawrence Livermore National Laboratory, 2024.
- **S. Kamboj**, "Simulating Quantum Systems, and Quantum Key distribution." Invited talk, Friday Seminar Series, CSU Stanislaus, 2024.
- **S. Kamboj** and H. M. Hurst, "Topology of a Driven 2D Spin Torque Oscillator Array." APS March Meeting, Chicago, 2022. Abstract T00.249.

Scientific Software and Projects

nonmarkov_qec
Open-source Python package; released on PyPI *2026 – Present*

- Built a surface-code benchmarking pipeline combining stochastic noise generation, Stim simulation, PyMatching decoding, and Monte Carlo sampling with bootstrap uncertainty estimates.
- Compared temporally correlated dephasing with a statistically matched white-noise baseline to distinguish noise-correlation effects from changes in marginal error strength.
- Maintained automated tests, strict type checking, continuous integration, and automated package releases, with documented methods and reproducible benchmark scripts.

Physics Concept Analyzer Tool (PCAT) UC Merced
Developer; Python, Flask, Gemini API, Google Cloud *2025*

- Designed and deployed a physics-teaching application on Google App Engine that guides students through concept identification, assumptions, and model construction before calculation.

Technical Skills

Theoretical and computational physics: Hamiltonian modeling; light–matter and magnon–photon interactions; spin-torque oscillator arrays; normal-mode and Bogoliubov analysis; symmetry-resolved spectra; sparse exact diagonalization; open-system dynamics; numerical convergence analysis.

Data analysis and visualization: Linear-inversion process tomography; pseudo-inverse reconstruction; Ramsey and relaxation-time fitting; Monte Carlo simulation; bootstrap uncertainty estimates; spectral, state-space, and parameter-map visualization with Matplotlib.

Scientific programming: Python, NumPy, SciPy, QuTiP, Matplotlib, Stim, PyMatching, Qiskit; Git, Linux, Bash, SLURM, pytest, ruff, mypy, and LaTeX.

Device simulation and control: COMSOL RF and AC/DC; electromagnetic parameter extraction; superconducting-qudit pulse control and process characterization.

Teaching and Mentoring

University of California, Merced and San Jose State University California
Teaching Assistant, Physics *2021 – Present*

- Led laboratory and discussion activities, held office hours, and assessed student work; connected measurements and uncertainty to physical models in introductory laboratories for biological science students.
- Collaborated with faculty on inquiry-based laboratory exercises and piloted PCAT to support students' reasoning during problem solving.

Guest lecturer, Computational Physics (PHYS 281), UC Merced 2024, 2026

Honors and Awards

Outstanding Graduate Teaching Award, UC Merced	2026
Physics Graduate Group Summer Fellowship, UC Merced	2025, 2026
Physics Graduate Group Travel Fellowship, UC Merced	2026
Outstanding Teaching Assistant Award, San Jose State University	2022
First place, Tech-Invent competition; silicon particle detector presentation	2017

Service and Outreach

Physics Delegate, Graduate Student Association, UC Merced	2024
Member, Graduate Dean Advisory Council on Diversity, UC Merced	2023
Mentor, Learning Assistant Reflection, Guidance and Exploration program	2022